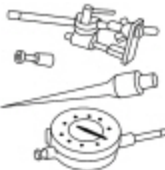
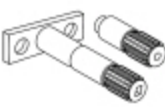


Differential Case and Ring Gear — Conventional



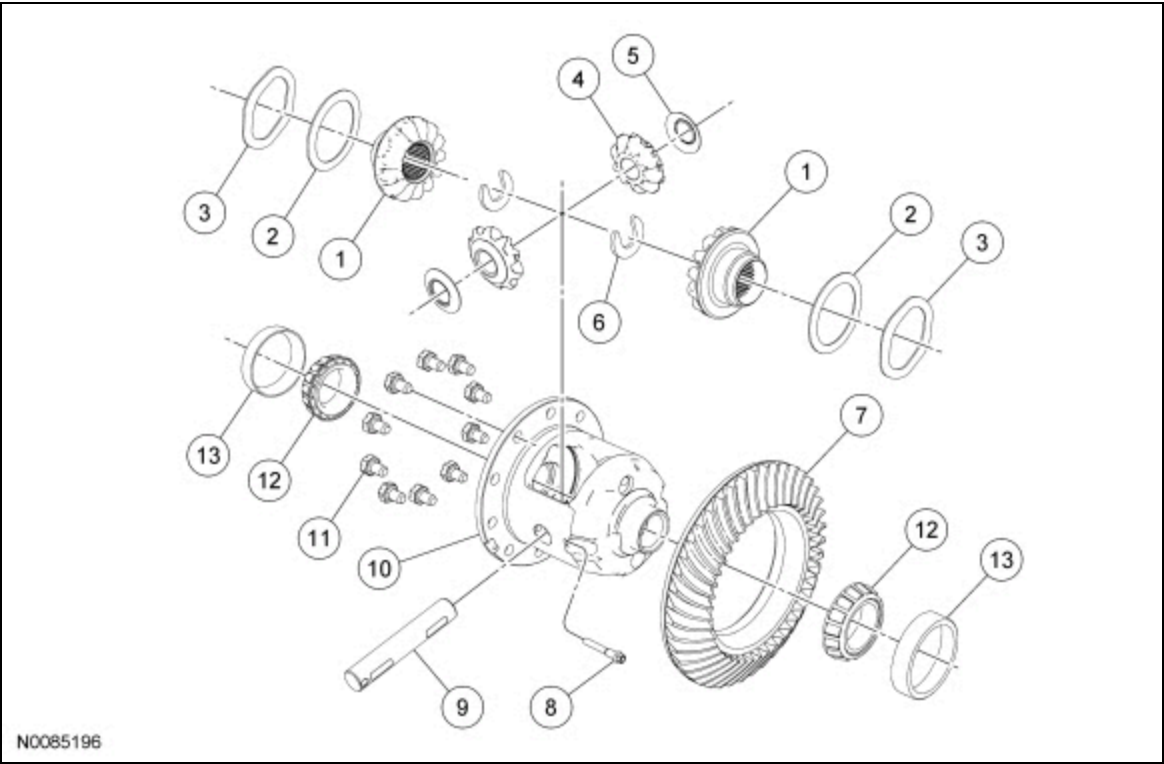
Special Tool(s)

 ST1214-A	Dial Indicator Gauge With Holding Fixture 100-002 (TOOL-4201-C)
 ST1265-A	Gauge, Differential Clutch 205-022 (T66L-4204-A)
 ST1749-A	Gauge, Differential (Traction Lock) 205-386 (T97T-4205-D)
 ST1375-A	Installer, Differential Side Bearing 205-010 (T57L-4221-A2)
 ST1220-A	Remover, Differential Bearing 205-116 (T77F-4220-B1)
 ST1543-A	Step Plate 205-D016 (D80L-630-5) or equivalent

Material

Item	Specification
Motorcraft® SAE 75W-140 Synthetic Rear Axle Lubricant XY-75W140-QL (US); CXY-75W140-1L (Canada)	WSL-M2C192-A and GL-5
Motorcraft® Premium Long-Life Grease	ESA-M1C75-B

XG-1-E1	
Motorcraft® Threadlock and Sealer TA-25-B	WSK-M2G351-A5

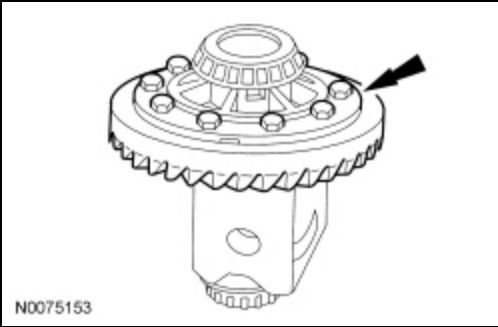


Item	Part Number	Description
1	4236 4236	Differential side gears
2	4228 4228	Differential side gear thrust washers
3	—	Differential side gear wave washers
4	4215 4215	Differential pinion gear (2 required)
5	4230 4230	Differential pinion thrust washer (2 required)
6	4N237 4N237	Axle shaft U-washer (2 required)
7	4209 4209	Differential ring gear
8	4241 4241	Differential pinion shaft lock bolt
9	4211 4211	Differential pinion shaft
10	4204 4204	Differential case
11	4216 4216	Differential ring gear bolt (10 required)
12	4221 4221	Differential bearings
13	4222 4222	Differential bearing cups

Initial disassembly

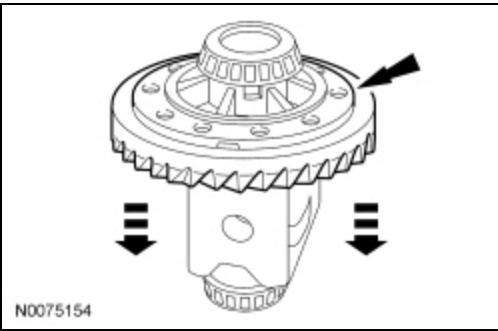
NOTE: If the differential ring gear backlash between teeth is greater than specification, remove the ring gear to check the differential case flange runout. Do not disassemble the differential carrier assembly.

1. Remove the differential carrier. Tag the differential shims and bearing cups for location. For additional information, refer to Differential Carrier in this section.
2. Remove and discard the 10 differential ring gear bolts.



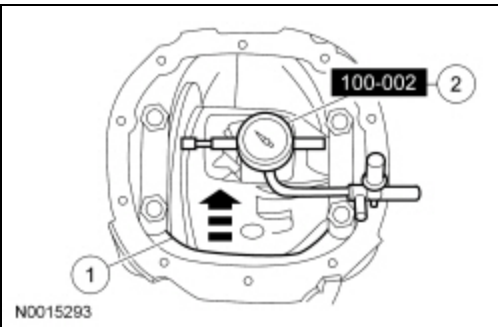
3. **NOTICE:** Care should be taken not to damage the differential ring gear bolt hole threads.

Insert a punch in the differential ring gear bolt holes and drive the differential ring gear off.



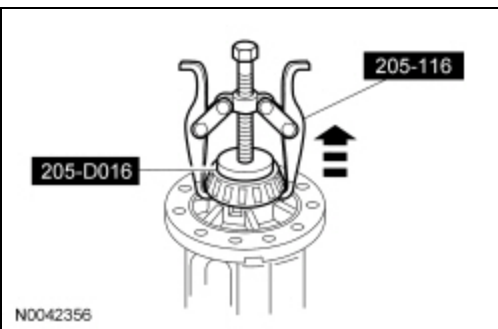
4. If checking differential case flange runout, install the bearing cups in their original location and install the differential assembly.

1. Install the bearing caps and differential shims.
 - Rotate the differential assembly to verify the bearings have seated.
2. Install the Dial Indicator Gauge With Holding Fixture and record the differential case runout.

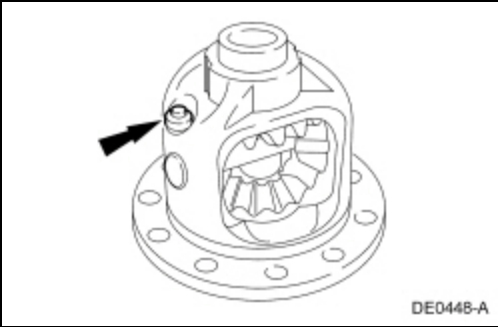


Final disassembly

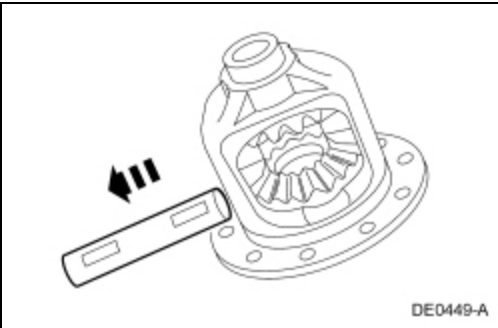
5. Using the Differential Bearing Remover and Step Plate, remove the differential bearings.



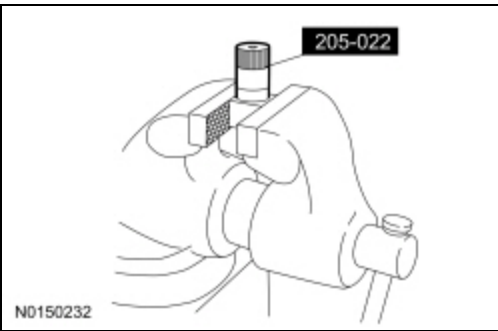
6. Remove the differential pinion shaft lock bolt.



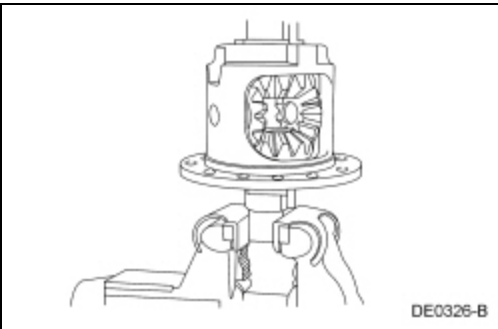
7. Remove the differential pinion shaft.



8. Install the Differential Gauge in a suitable vise.

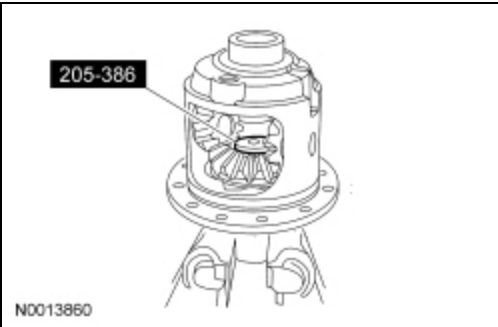


9. Install the differential case on the Differential Gauge.

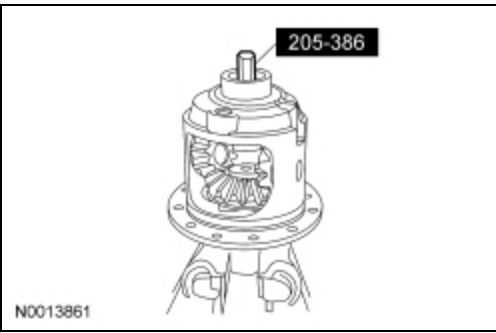


10. **NOTE:** Apply a small amount of grease to the centering hole of the Differential Gauge.

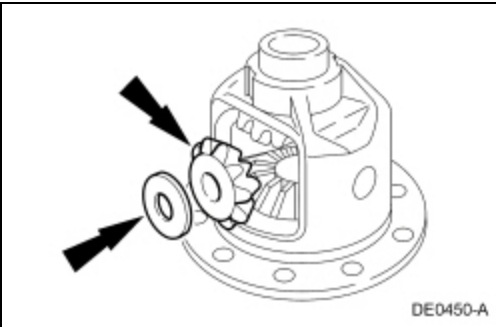
Install the Differential Gauge step plate in the bottom differential side gear.



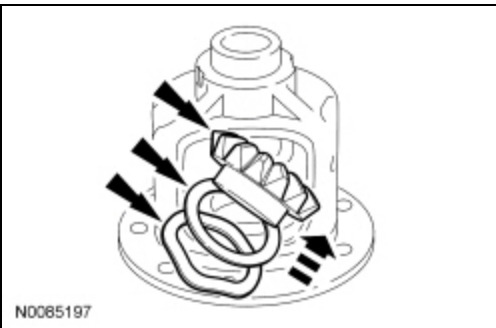
11. Install the Differential Gauge nut in the upper differential side gear. Hold the nut in position while installing the Differential Gauge hex screw. Tighten the hex-head screw until contact is made with the Step Plate.



12. Rotate and remove the differential pinion gears and differential pinion gear thrust washers.



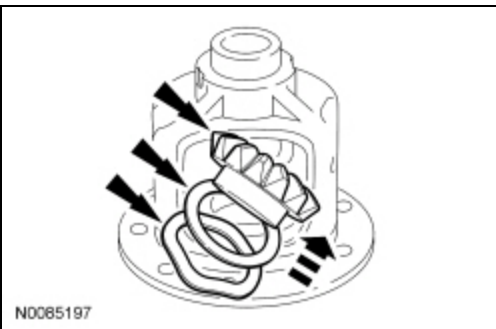
13. Remove the differential side gears, the differential side gear washers and the differential side gear wave washers.



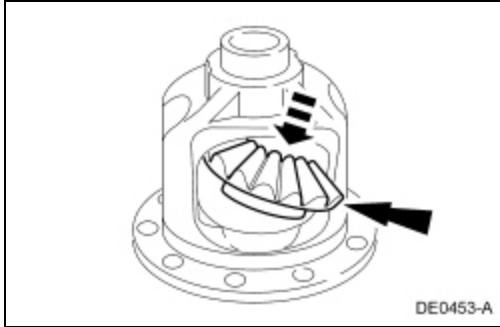
Assembly

1. **NOTE:** Lubricate the differential side gear thrust washers, wave washers and the side gear journals with axle lubricant.

Position the differential side gear thrust washers and wave washers on the differential side gears.



2. Install the differential side gears.

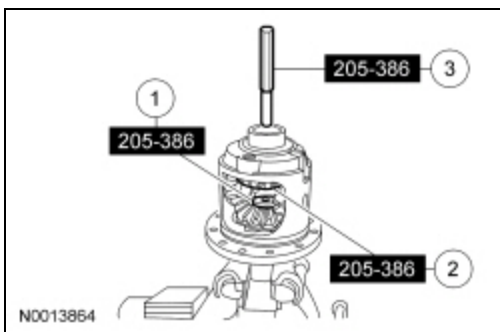


3. **NOTE:** Apply a small amount of grease to the Step Plate bore.

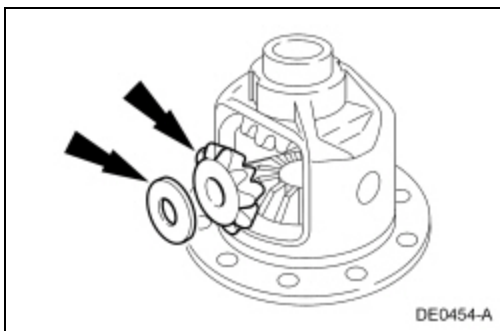
NOTE: If necessary, insert the Differential Gauge dowel bar in the nut bore to keep the nut from turning as the hex screw is tightened.

Assembly the Differential Gauge to the differential case.

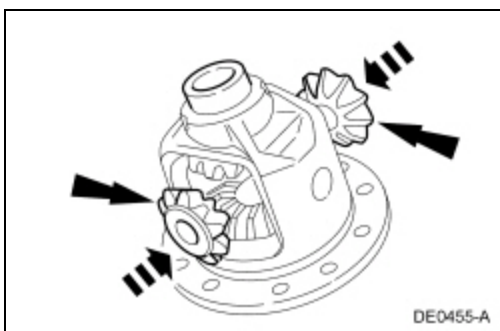
1. Position the Differential Gauge step plate in the bottom differential side gear.
2. Position the Differential Gauge hex screw plate in the top differential side gear and hold it in place.
3. Install the Differential Gauge hex screw and tighten it 2 turns after it contacts the bottom Step Plate.



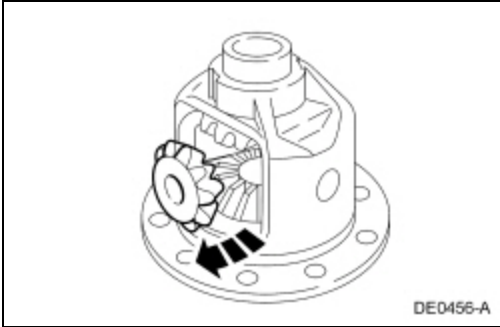
4. Use axle lubricant to lubricate the differential pinion gear thrust washers and the differential pinion gears.



5. Install the differential pinion gears and differential pinion gear thrust washers opposite the differential side gears.

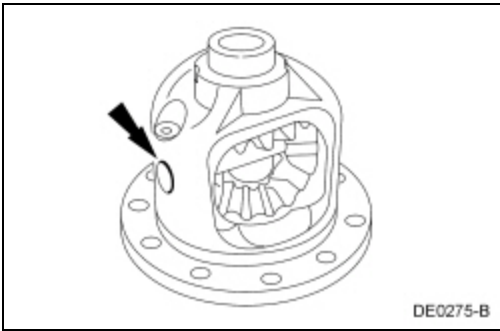


6. Rotate the differential pinion gears to align with the differential pinion shaft bore.



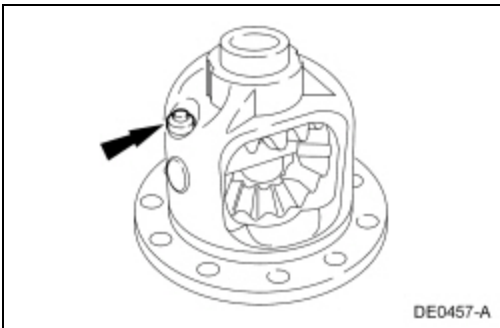
7. Install the differential pinion shaft.

- Align the hole in the differential pinion shaft with the hole in the differential case.

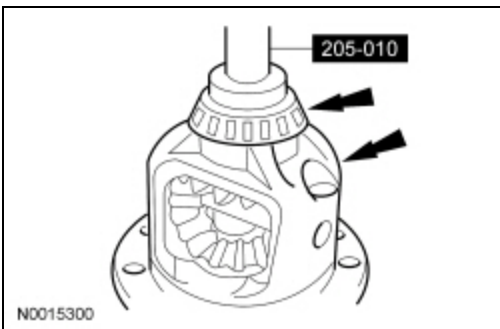


8. **NOTE:** If a new pinion shaft lock bolt is unavailable, coat the threads with threadlock and sealer prior to installation.

Install a new differential pinion shaft lock bolt and tighten finger-tight.



9. Using the Differential Side Bearing Installer, install the new differential bearings on the differential case.

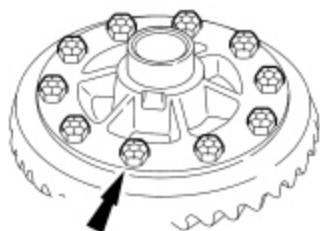


10. Position the differential ring gear on the differential case. Align the bolt holes by starting 2 bolts through the holes in the differential case and into the differential ring gear.

11. Using a shop press, press the ring gear on the differential case.

12. Install the 10 new differential ring gear bolts. Tighten in 2 stages.

- Stage 1: Tighten to 60 Nm (44 lb-ft).
- Stage 2: Tighten an additional 90 degrees.



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13. Install the differential carrier. For additional information, refer to Differential Carrier in this section.